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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gangopadhyay, Anup Ph.D. (Washington University in St. Louis)		
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	This grant is a continuation of previous grants NNX10AU19G and NNX16AB52G. Funding is provided for Prof. Dr. Kenneth Kelton and Dr. Anup Gangopadhyay for their role in incorporating all ground-based studies and flight preparation activities by the QUASI team at Washington University in St. Louis (WUSTL). These support QUASI flight investigations on the International Space Station (ISS) and collaborative flight projects with ESA teams ICOPROSOL (scientific coordinator – Dr. Dirk Holland-Moritz, DLR) and THERMOPROP (scientific coordinator – Prof. Dr. Hans Fecht, Ulm University). The flight experiments on the ISS are in designated batches. Batch 1 and batch 2 experiments have now been successfully completed. The planned activities under this grant are for samples approved for batches 3 and 4. Metallic alloy liquids processed in a containerless environment are investigated in all of the proposed measurements for		

Task Description:

QUASI, THERMOPROP and ICOPROSOL. The primary goals of the flight and ground-based experiments are:

1. An investigation of the role of stirring and diffusion on crystal nucleation and a comparison of the experimental results with predictions of a new model for nucleation, the Coupled Flux Model;

2. Measurements of key parameters needed for a quantitative analysis of the nucleation data obtained in the ground-based and ISS studies (driving free energy, specific heat and viscosity);

3. Measurements of the electrical resistivity of metallic liquids as a function of temperature to investigate a proposed correlation between structural changes in the liquid and dynamical properties, and to confirm the existence of a plateau in the temperature dependence of the resistivity that we have recently discovered at high temperatures;

4. Measurements of the thermal conductivity of metallic liquids at high temperature;

5. Measurements of the crystal growth velocity in the ground-based and ISS studies.

Rationale for the Research

Measurements of the maximum supercooling of free-cooling metallic liquids that crystallize to phases of a different composition than the liquid, are made in the quiescent environment of the electrostatic levitation facility (WU-BESL) located in our laboratory at Washington University in St. Louis. Long-range diffusion is required for nucleation in these cases. These ground-based results are compared with undercooling studies made as a function of stirring in the ISS-EML. According to the Coupled Flux Model for nucleation, developed by Kelton, the nucleation rate should increase with the stirring rate, since long-range diffusion will become less important. A key goal of our ongoing research is to investigate this.

Measurements of the thermophysical properties, including the specific heat, density, surface tension, viscosity, thermal conductivity and electrical resistivity, will be made using the WU-BESL and the ISS-EML, since these facilities have different capabilities for these studies. The specific heat, density, surface tension and viscosity are needed for a quantitative analysis of the nucleation data. Since our earlier studies have demonstrated that ordering in the liquid couples to the nucleation barrier, the undercooling and thermophysical property measurements will be correlated with x-ray synchrotron diffraction results for the supercooled liquids of interest, obtained previously using WU-BESL. Based on our previous studies, the specific heat and electrical resistivity are also valuable for providing structural evidence for a cooperative cross-over behavior in the shear viscosity at high temperatures. When coupled with the measured electrical resistivity, measurements of the thermal conductivity will allow an investigation of the Widemann-Franz law for high temperature metallic liquids, a case where it is not been studied before.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which have the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes, cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. The combination of the containerless based methods of WU-BESL and the ISS-EML enables these measurements. In addition to their relevance for processes such as additive manufacturing on earth, where nucleation in highly stirred conditions is important, measurements in two different gravitational fields will be important for designing additive manufacturing in space and on the Moon. Finally, past measurements of the electrical resistivity in high temperature liquids have revealed a plateau in the temperature dependence of the resistivity at temperatures near a crossover temperature for the viscosity. This is not predicted by any existing theory. Our measurements will provide valuable information for developing a theoretical understanding of this phenomenon.

A summary of activities carried out during year 2 of this grant is listed below.
1. Measured the specific heat and thermal conductivity in liquid Zr80Pt20 as a function of temperature using the modulation technique available on the International Space Station (ISS) electromagnetic levitator (EML) facility. The magnitudes of both quantities show a decrease with increasing temperature. The specific heat, however, shows a maximum near 1400K; the source for this is not yet clear. By combining the thermal conductivity measurements with the electrical resistivity measurements discussed in last year's report, the validity of the Widemann-Franz (WF) law for metallic liquids at high temperatures can be investigated. This law predicts that the ratio of the thermal conductivity to the electrical conductivity is equal to a universal constant multiplied by the temperature in absolute units. Surprisingly, the measurements show that it is violated in this case, not only for Zr-Pt, but also for two other liquids that we studied in batch 1 and batch 2 experiments. This has serious practical consequences since the WF law is frequently used to estimate the thermal conductivity in liquids, since it is difficult to measure on Earth due to the contribution of convective flow to the thermal current. Fundamentally, our research opens up new areas of needed research to understand the reasons for the violation. Our studies show that the WF law fails when the electrical resistivity of the metallic liquid is high, suggesting that the breakdown is due to the high level of disorder, causing a decoupling between charge and heat transport.

2. Two additional batch 3 samples, Cu47Zr47Al6 and Ti45Zr45Ni10, will be studied on the ISS to explore the effects of stirring on nucleation. As a baseline under quiescent conditions, we have studied these under terrestrial conditions using our electrostatic levitation (ESL) facility, Washington University's Beamline Electrostatic Levitator (WU-BESL). As will be the case for the ISS data, the ESL data have been analyzed using the Skripov method, where the maximum supercooling data are fit to a Poisson probability distribution as a function of temperature. For good statistics, 80 supercooling cycles were made for the Cu47Zr47Al6 liquid; it was possible to obtain 180 cycles for the Ti45Zr45Ni10 liquid. The fits show that the interfacial free energy for Cu47Zr47Al6 is slightly larger than for Ti45Zr45Ni10, most likely reflecting a greater difficulty to nucleation the more complex crystal phase in Cu47Zr47Al6.

Task Progress:	3. Recently, questions of whether there exists a lower bound in the viscosity that is related to quantum mechanics have arisen within a diverse community, including quantum gravity, high energy physics and condensed matter physics. Surprisingly, this has escaped the attention of materials scientists and the fluid dynamics community, perhaps because of the general belief that quantum mechanics is of little consequence for classical fluid dynamics. Other researchers, looking at the hydrodynamic behavior of a special type of black holes called black branes, have suggested that the ratio of the shear viscosity and the entropy density (s) has a universal lower bound that is determined by Plank's constant, h and Boltzmann's constant, kB. Using the measured viscosity and specific heat data (to calculate the entropy density) for two liquids, Vit 106 (Zr57Nb5Cu15.4Ni12.6Al10) and Ti39.5Zr39.5Ni21, we find that the ratio of the viscosity to the entropy density of the experimental data are in agreement with the theoretical predictions to within approximately one order of magnitude, suggesting that the theoretical predictions are valid even for classical fluids. The fact that quantum mechanics may play such an important role at high temperatures in the liquid viscosity and that so many different systems follow the same lower bound is fundamentally surprising. 4. The Gruenisen parameter is usually determined from the temperature rise during adiabatic compression of a material. This parameter and the bulk modulus are needed to describe the equation of state of any material as a function of pressure and temperature. Through a combination of ISS and ground-based measurements, we have determined the magnitude and temperature dependence of the Gruenisen parameter and the bulk moduli for liquids and of the bulk modulus of a bulk metallic glass former near the glass transition temperature. An analysis of the data shows that a simple rule of mixing estimate for the bulk modulus can be made using elemental data. Using those estimates and the experimental data for the volume expansion coefficient and the specific heat, which are directly measurable, the Gruenisen parameter can be obtained at room temperature. Our measurements have also shown that the temperature dependence of the Gruenisen parameter is small and can be treated as temperature independent. This provides the first method to make reasonable estimates of this important property for liquids. 5. Anup Gangopadhyay, Co-Investigator (Co-PI) on this grant, will support an in-person Investigator Working Group (IWG) meeting that will be held May 23 – 24, 2023 at the German Space Agency (Deutsches Zentrum für Luft- und Raumfahrt/DLR). 6. Principal Investigator (PI) Kelton has completed and submitted an invited perspective entitled, "A Perspective on Metallic Liquids and Glasses," to the Journal of Applied Physics. The measurements of the electrical resistivity of Zr80Pt20 made on the ISS under the support of this grant play a key role in that article. These accomplishments are described in greater detail in the annual technical document submitted to NASA.
Bibliography Type:	Description: (Last Updated: 05/20/2025)
Articles in Peer-reviewed Journals	Gangopadhyay AK, Nussinov, Z, Kelton, KF. "Quantum mechanical interpretation of the minimum viscosity of metallic liquids." Phys. Rev. E. 2022 November 22;106(5):054150. http://dx.doi.org/10.1103/PhysRevE.106.054150 , Nov-2022
Articles in Peer-reviewed Journals	Kelton KF. "A perspective on metallic liquids and glasses." J. Appl. Phys. 2023 Jul 7;134(1). https://doi.org/10.1063/5.0144250 , Jul-2023